

LAUNCHPAD

Robotics Team Operations System



Launchpad is an offline-first, client-side browser management application designed to coordinate parts inventories, bills of materials (BOM), vendors, tooling checkouts, member roles, kanban tasks, and workspace organization for high-performing engineering and robotics teams.

PRODUCT DIRECTORY

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Status: Deployed & Active

CORE PLATFORM & ARCHITECTURE

- **Offline-First Storage:** Driven entirely in the browser using IndexedDB to store data locally. Requires zero server configuration, runs standalone, and works with standard file schemes or local static servers.
- **Data Portability (Backup/Restore):** Allows teams to backup the entire application state into a single portable JSON file, and restore it on another machine to prevent data loss or sync team workspaces.
- **PIN-Based Login & RBAC:** Incorporates role-based access controls supporting Captains, Leads, Mentors, and Members. Restricts moderation features to authorized users using secure 4-digit PIN authentication.
- **Dynamic Responsive Shell:** A custom UI dashboard framework crafted in vanilla CSS that automatically collapses and fits mobile screens, ensuring compatibility on laptops, tablets, and phones.
- **Secret Cat Command:** A hidden developer shortcut (Cmd + Shift + Option + K) that instantly spawns an animated cat overlay in the top-right corner of the interface.

PROJECTS & SUB-PROJECTS MODULE

- **Hierarchical Structure:** Supports creating top-level projects that can contain nested sub-projects to keep complex robotics assemblies and events organized.
- **Progress Analytics:** Automatically calculates project completion progress dynamically based on the percentage of completed tasks linked to the project.
- **Advanced Template Duplication:** Clones existing projects, replicating the exact hierarchy, linked BOM parts, and tasks while resetting statuses to default parameters for new builds.
- **Consolidated Detail Tabs:** Dedicated project detail dashboards presenting nested views for linked sub-projects, Bill of Materials items, and assigned tasks in one tabbed view.

PARTS CATALOG & INVENTORY MODULE

- **Spreadsheet Table View:** Displays parts in a sortable grid showing photos, name, category, vendor, location, stock level, needed quantity, and unit cost.
- **Smart Restock Filters:** Automatically highlights parts that have dropped below safety stock quantities (In Stock < Needed) and allows location or category filtering.
- **Quick Stock Adjuster:** Features an incremental modifier modal to easily log parts received or consumed in bulk (e.g. +10, -5) without opening full edit records.
- **Media Attachment:** Allows file uploads to capture photos of parts, storing them directly in the IndexedDB catalog as high-fidelity Base64 DataURLs.
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Bulk Operations: Allows users to select multiple items via checkboxes and execute bulk deletion tasks in a single click.

BILL OF MATERIALS (BOM) MODULE

- **Per-Project BOM Trackers:** Links parts catalog records directly to projects, tracking procurement and installation requirements for specific designs.
- **BOM Line Item Totals:** Automatically computes unit costs, needed quantities, total line costs, and sums up the overall project cost in a dedicated summary footer.
- **Procurement Workflow:** Tracks parts through four distinct stages: Not Started, Ordered, In Stock, and Installed.
- **Stat Cards & Progress Bars:** Renders high-level summary cards showing Total Items, Total Budget, Installed count, and percentage progress indicator.
- **CSV Data Export:** Enables one-click CSV compilation and download, formatted to import directly into Excel or Google Sheets for team budgeting.

TOOL CHAIN & CHECKOUT MODULE

- **Tool Directory:** Maintains an active log of tools, showing photos, storage locations, and active checkout states.
- **Condition Assessment:** Visual badges indicating tool health: Good (green), Needs Maintenance (amber), and Broken (red).
- **Interactive Checkout Engine:** Enables members to check out tools by selecting their name from the team roster. Tool cards dynamically switch to reflect checkout status.
- **Visual Search & Filter:** Allows searching tools by name or filtering by availability (Available vs Checked Out).

TEAM ROSTER & PEOPLE MODULE

- **Role Categorization:** Tracks members categorized as Captain, Mentor, Lead, or Member with corresponding custom avatar icons.
- **Activity Tracking:** Detailed profiles show all active tasks assigned to the member and all tools currently checked out under their name.
- **Access Moderation Panel:** Allows Mentors to approve pending user registrations before they are granted read/write permissions to the database.

KANBAN TASK BOARD

- **Interactive Kanban Tracks:** Tracks tasks visually across three workflow columns: To Do, In Progress, and Done.
- **Metadata Overlay:** Kanban cards display title, project association, assignee avatar, due dates, and priority indicators.

Priority Visualizers: Color-coded priority dots signifying Low (green), Medium (orange), and High (red) priorities.

- **Advanced Roster Filtering:** Allows filtering task boards dynamically by assignee or by specific projects.

WORKSPACE & STORAGE MAP

- **Interactive 2D Floorplan:** Renders a visual 600px floorplan mapping storage bins, workspaces, and machinery.
- **Coordinate Rendering:** Uses percentage-based coordinates (X, Y position and W, H dimensions) to scale and draw zones accurately on any screen size.
- **Location Audit Sidebar:** Clicking any map zone opens a details panel listing all parts and tools currently registered at that location.